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Palaeomagnetism of early Cambrian Brigus Red Shale, eastern Newfoundland

Maurice K. Seguin and Kunduri V. Rao *

Paleomagnetic Laboratory, Faculty of Science and Engineering, University of Laval, Quebec City. P.Q. G1K 7P4 (Canada)

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Studies of the early Cambrian Brigus (BR) red shale exposed in Conception Bay yielded a single, stable mean remanence direction of $D = 265°$, $I = +58°$ (reverse polarity) before tilt correction and $D = 279°$, $I = +48°$, $k = 47$, $\alpha_{95} = 6.9°$ for $N = 7$ sites (25 samples) after correction for gentle tilt; the corresponding (North pole palaeopole) position is 27° S, 51° E (dp, dm = 6°, 9°). This magnetization is characterized by a linear vector trajectory at higher unblocking temperature (T_{UB}) spectra (400–680° C). The magnetization directions at the low T_{UB} range (400° C) are more variable from sample to sample, although they typically have steep downward inclinations, with no evidence of Kiaman overprinting. Marginal but not statistically significant improvement ($k = 44$–77) in the grouping of site means after tilt correction suggests that the predominant, univectorial westerly remanance was probably acquired after deposition.

Based on both geological and magnetic evidence, the Brigus Shale Formation appears to be remagnetized during Siluro-Devonian times. The position of the far-sided antipole BR (27° N, 129° W) located in the vicinity of the pole for the Cambrian Morrison River Formation (MR at 20° N, 146° W), which is discordant from the younger Palaeozoic (Late Ordovician–Carboniferous) poles reported for the Avalon Zone, is fortuitous. Since the magnetization of BR is most likely secondary, the magnetization is younger than the Iapetus closure date determined geologically (Late Ordovician). Consequently, the displacement of the Avalon Zone with respect to the North American craton cannot be estimated; the width of the Proto-Atlantic Ocean cannot be calculated.

1. Introduction

The Avalon Zone (Williams, 1979), well exposed in the Avalon Peninsula of eastern Newfoundland, consists mainly of thick sequences of Upper Precambrian igneous and sedimentary rocks of complex stratigraphic interrelationships. These sequences are locally overlain by transgressive Cambrian and Lower Ordovician shallow water platformal sedimentary rocks with body and trace fossils typical of the Atlantic faunal realm (King, 1982). In a proposed model for the tectonic evolution of the Appalachian Orogen, the Avalon Zone is considered to be a suspect terrane (Williams and Hatcher, 1982) with its Proterozoic and earliest Palaeozoic development different from that of the neighbouring Gander and Meguma terranes. O'Brien et al. (1983) have proposed a tectonic model for the Proterozoic evolution of the Avalon Terrane, analogous to those of Pan-African belts, and in which Avalon became juxtaposed with the Appalachian orogen in early Palaeozoic time.

Brückner (1980) discusses the stratigraphic development of the Avalon Peninsula in terms of three major 'assemblages' developed successively in fundamentally different palaeogeographic set-

* Present address: Haileybury School of Mines, Haileybury, Ontario, POJ 1KO Canada.

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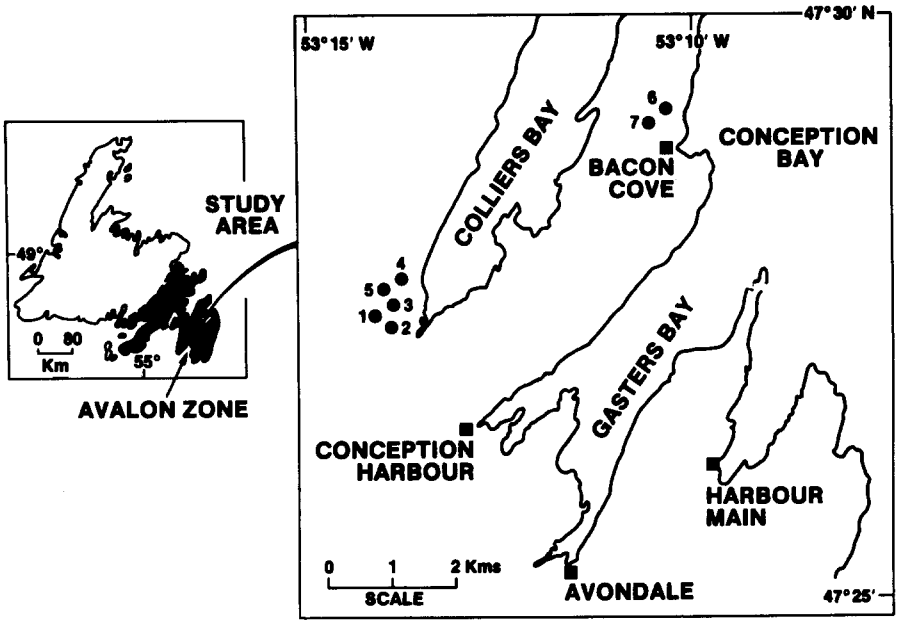


Fig. 1. Distribution of palaeomagnetic sampling sites (solid circles) in southeastern Conception Bay, Avalon Peninsula, Newfoundland. For general geology and location of principal known faults of the area refer to fig. 2 in McCartney (1967).

TABLE I

Magnetization in the Cambrian Brigus red shale, Conception Bay (Eastern Newfoundland) [a]

NRM				Thermal					
Site	$N_1(n)$	In situ $D°, I°$	k	$N_2(n)$	T_{UB} Range [b]	In Situ $D°, I°$	Tilt-corrected $D°, I°$	k	α_{95} (°)
BR-1	5(10)	295, +74	24	5(7)	400–680 °C	281, +53	286, +37	48	11.7
BR-2	5(12)	256, +55	11	5(8)	550–680 °C	267, +57	274, +42	56	10.3
BR-3	5(8)	257, +82	53	3(4)	500–680 °C	252, +69	276, +57	66	11.6
BR-4	5(15)	204, +78	34	2(4)	400–650 °C	272, +67	278, +49	–	–
BR-5	5(8)	273, +65	17	4(6)	500–660 °C	241, +63	261, +53	61	11.8
BR-6	4(14)	231, +39	26	3(5)	400–680 °C	267, +50	288, +47	88	14.9
BR-7	3(8)	262, +33	18	3(5)	400–660 °C	268, +40	288, +46	78	10.2

	N	$D°$	$I°$	R	k	α_{95}	α_{99}	Latitude	Longitude	dp, dm	Pole [c]
Thermal, *In situ*	7	265.5	+57.6	6.8626	44	9.2	12.3	25 °N,	113 °W	10,14	BR′
Thermal, tilt-corrected	7	279.3	+47.6	6.9218	77	6.9	9.3	27 °N,	129 °W	6,9	BR

[a] N_1 and N_2 are number of samples used in statistics for NRM and after thermal treatment, respectively: n is the number of specimens used in each case; $D°$, $I°$ are declination and inclination of remanent magnetization; k is the estimate of Fisher's (1953) precision parameter; α_{95} is semi-angle of cone of 95% confidence for mean direction; R is the resultant of N unit vectors.

[b] Range of unblocking temperature, T_{UB}, over which the directions are computed from linear trajectories in Zijderveld (1967) vector diagrams.

[c] Symbol used for the corresponding magnetic south pole (Fig. 4).

tings. While the 'Lower' and 'Middle' assemblages are generally considered Hadrynian in age, the fossils in the 'Upper' assemblage indicate a wide age bracket ranging from earliest Cambrian to early Ordovician. The 'Upper' assemblage is further subdivided into four stratigraphic sequences, the lower two being the Adeyton and Harcourt Groups of Cambrian age (Hutchinson, 1962; Jenness, 1963). The oldest fossiliferous beds in the Adeyton Group form the Bonavista Bay and Smith Point formations, consisting of red and pink slate and limestone. The overlying Brigus Formation contains red shale, slate and minor amounts of limestone. These Cambrian formations commonly occur as synclinal outliers or down-faulted remnants surrounded by large areas of Precambrian rock. Considering the large angular unconformities between the formations of differing age, the establishment of a stratigraphic correlation chart is difficult.

The present palaeomagnetic study is based on the Brigus Formation exposed in two small synclinal outliers in southeastern Conception Bay (Fig. 1) where the Cambrian sedimentary rocks are relatively undeformed and unmetamorphosed. Hutchinson (1962) assigned a medial to late Lower Cambrian age to the Brigus Formation based on faunal evidence. Detailed accounts of the stratigraphic and structural development of the Brigus Formation are presented elsewhere (Hutchinson, 1962; McCartney, 1967, 1969). Palaeomagnetic sampling was carried out mainly at two localities: one (sites 1–5) just west of Colliers Bay where the beds have a shallow (20°) and uniform dip to the northwest; and the other (sites 6 and 7) near Bacon Cove where the Brigus rocks dip eastwards. A total of 32 oriented hand samples were collected from seven sites (with a minimum of three to a maximum of five samples per site) with the help of a Brunton compass. The Cambrian and Ordovician sediments in southeastern Conception Bay have a shallow and uniform geologic dip, probably acquired in post-early Ordovician times.

2. NRM results

The natural remanent magnetization (NRM) was measured with a digital Schonstedt spinner magnetometer (DSM-1) and the step-wise alternating field (AF) and thermal demagnetizations were carried out, respectively, with Schonstedt GSD-1 and TSD-1 units. The directions of the NRM are characterized by westerly declinations, intermediate to steep downward inclinations and pronounced deviations from the present-day Earth's field (PEF) at the sampling sites ($D = 334°$, $I = +71°$). This suggests that the samples are palaeomagnetically suitable (Table I, Fig. 2A). Directions far from the PEF may simply reflect strong, stable ancient but still secondary, magnetizations. For instance, the middle member of the Cambrian Wood Canyon Formation (Gillett, 1982), in Southern Nevada, has NRMs far from the present field, streaked toward a Kiaman-like direction and it was proven that this formation was remagnetized in the late Palaeozoic. In the present study, the site directions could have been contaminated by the PEF. The NRM intensities

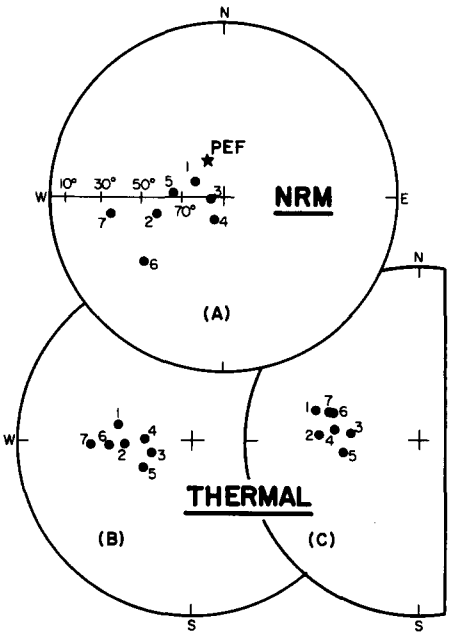


Fig. 2. Site-mean direction of magnetization (A) for NRM without tilt correction; (B) after thermal treatment and before tilt correction; (C) after thermal treatment and tilt correction. Directions are plotted on a Schmidt net. Solid circles represent projections on lower hemisphere. The star represents the present Earth's field (PEF). The sites are identified with their corresponding numbers (Fig. 1, Table I).

range over an order of magnitude from 2 to 20 × 10^{-3} A m^{-1}.

3. Demagnetization

Laboratory treatment of the oriented samples involved standard procedures of step-wise AF and thermal demagnetization and subsequent analysis with Zijderveld (1967) diagrams and vector subtraction. Demagnetization with AF treatment up to a maximum of 100 mT was not effective; the directions did not move from their NRM values. Sometimes, spurious magnetizations took place in higher demagnetizing fields. Consequently the whole collection was analysed by thermal treatment alone. Thermal treatment isolated a characteristic remanence by the 500 °C step, which subsequently unblocked at just below 660 °C. Generally, above the 400 °C step, the demagnetization trajectories are linear towards the origin until the magnetization drops below noise level by the 660 °C step, as shown for two representative specimens (Fig. 3). This last trajectory with high unblocking temperature (T_{UB} = 500–680 °C for BR1-4C and T_{UB} = 400–660 °C for BR7-5B, Fig. 3) is interpreted as the removal of a single remaining magnetization ($D = 280°$, $I = +48°$), which, on the basis of a similar direction from sample to sample, can be considered the predominant magnetic signature of the Brigus red shale. The directions generally become erratic after the 660 °C step and in some cases the remanence could not be measured owing to weak magnetization in the samples. The magnetizations removed at low-temperature steps (T_{UB} range 400 °C) are more variable in direction from sample to sample, although they typically have steep downward inclinations. The low demagnetization temperatures are not the result of spurious components from drill press magnetization, first because stainless-steel drill bits were not commonly used and second because there is no correlation between sample orientations (drilling axis is vertical) and the spurious directions (inclinations: 50–70°). No evidence of directions corresponding to Kiaman overprinting (Roy and Morris, 1983; Irving and

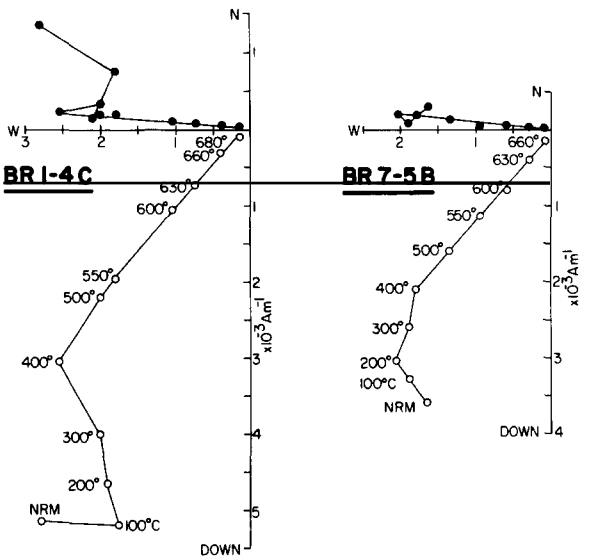


Fig. 3. Representative orthogonal thermal demagnetization diagrams (Zijderveld, 1967) of successive end-points of the magnetization vector for two specimens, as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps (°C). Units indicated for axes are in 10^{-3} A m^{-1}.

Strong, 1984, 1985) is observed during thermal treatment.

Detailed thermal demagnetizations were carried out for each site and the site-mean directions were calculated based on linear vector trajectories (approximate T_{UB} range: 400–680 °C) for each sample (Fig. 2B, Table I). It is commonly observed that components in haematite overlap greatly in stability spectra, in that cleaned directions are difficult to identify if the entire demagnetization path is linear. Bends and curves in the path at least give a sense of what is being removed. The absence of bends suggests that the component(s) of magnetization has (have) not been removed, implying that the residual component is either secondary or unresolved. The site-mean directions after correction for gentle geological tilt (sites 1–5 westerly, sites 6 and 7 easterly dips) yield a marginal improvement in grouping ($k = 44$ to $k = 77$, Table I, Fig. 2B and C), but not statistically significant (McElhinny, 1964). Following McCartney's description of fault systems in southern Conception Bay (fig. 2, McCartney (1967)), the two sampling localities in the present study (Fig.

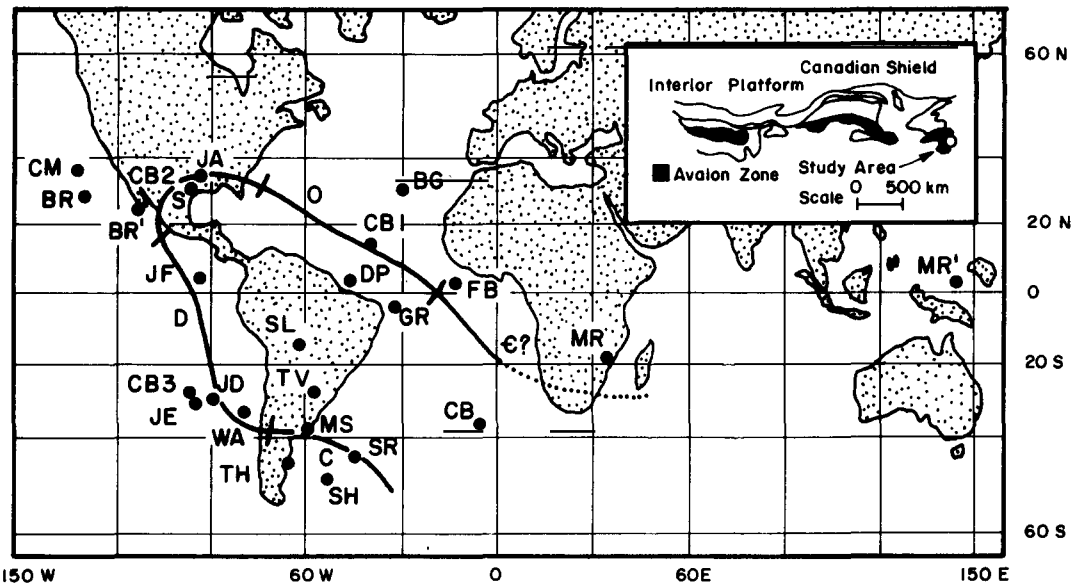


Fig. 4. Palaeomagnetic poles obtained from late Precambrian to Carboniferous rock units from the Avalon Zone (shaded area on inset). For pole symbols, see text and table 2 in Rao et al. (1986). Late Precambrian: GR, FB and CM (Irving and Strong, 1985). Cambrian: MR (Rao et al., 1986), CB (Rao et al., 1981), BG (Johnson and van der Voo, 1985) and possibly pole BR obtained in the present study. MR′ and BR′ are the respective poles obtained before tilt correction corresponding to MR and BR. Late Ordovician: DP (van der Voo and Johnson, 1985) and CB1 (Rao et al., 1981). Siluro-Devonian: JA (Roy et al., 1979), JF (Roy and Anderson, 1981), CB2 (Rao et al., 1981) and BR′ (this study). Early Devonian: JE (Roy and Anderson, 1981). JD (Roy et al., 1979), WA (Buchan and Hodych, 1982) and CB3 (Rao et al., 1981). Carboniferous: SL, SH, SR (Irving and Strong, 1985), TV, TH (Kent, 1982) and MS (Roy and Park, 1974). Mercator Projection. All poles are south poles.

1) belong to two different fault-bounded blocks, in which case it is possible that they have rotated with respect to each other during post-Middle Cambrian faulting. Consequently a simple tilt correction (a simple rotation about the present bedding strike) will not restore magnetization to original attitude, resulting only in a marginal improvement in the grouping of site means. Although the inclination would not be affected, an error may be introduced in the tilt-corrected declinations; however, absence of azimuthal smearing in the tilt-corrected site-mean directions (Fig. 2C) rules out such a possibility, implying the probable absence of local rotation.

In contrast to the steeply inclined NRM directions (in situ) at sites 1–5 (area west of Colliers Bay), the mean directions for sites 6 and 7 (near Bacon Cove) have intermediate inclinations (Figs. 1 and 2A and Table I). However, upon thermal treatment and tilt correction, the directions at sites 6 and 7 become indistinguishable from the rest of the sites (1–5), and this may be taken as evidence for the pre-tilt origin to the characteristic remanence $D = 279°$, $I = +48°$ (Table I). Uniformity of bedding attitudes and gentle geological dip of the sampling sites preclude a conclusive test on the age of acquisition and stability of magnetization in Brigus red shale units. Since there is no steep regional dip indicative of large regional deformation, the high-T_{UB} direction probably represents the magnetization acquired sometime during and/or after the Brigus shale deposition. Assuming that the magnetization ($D = 279°$, $I = +48°$; Table I) was acquired during or sometime after the deposition of the Brigus red shale, we calculated the corresponding palaeomagnetic (south) pole position which falls at 27° N, 129° W associated with an error oval dp, dm = 6°, 9° (Table I).

This antipole (BR) is compared with the published late Precambrian–Carboniferous (south) poles from the Avalon Zone (Fig. 4).

4. Magnetic memory

The remanence is present mainly, if not solely, in haematite. This conclusion is based on the following arguments. The persistence of the characteristic westerly remanence to a high T_{UB} range (400–680 °C) indicates a predominant contribution from haematite. Optical examination of polished sections shows that the most common opaque mineral is haematite with minor amounts of magnetite. The large spread (280 °C) in the high-T_{UB} spectrum during thermal treatment may suggest a minor contribution of magnetite. A variation of the grain size of haematite could also partly explain this spread. If magnetite were contributing to the magnetization, one would expect a glitch in the demagnetization curves as the Curie point of magnetite is crossed and this is not so. In addition, AF demagnetization would be partially efficient; although directions need not change much, one should see a decrease in intensity just from removal of a present VRM; this would be obvious on a vector diagram or J/J_0 plot, but it is not the case in this study.

5. Primary or secondary magnetization?

Even when the quality of the palaeomagnetic data appears to be good, as is the case in this study, the value of a study of this kind is incomplete if the arguments for a penecontemporaneous origin of the characteristic magnetization are not clearly put forward. Consequently, the validity of the interpretation of the results depends upon the determination of the origin of magnetization. When dealing with ancient rocks, it is quite safe to assume that they are remagnetized unless proven otherwise. However, in Palaeozoic or older rocks, the criteria required to arrive at a full proof from the evidence for a primary magnetization are not met in full. If data are lacking to constrain the age of magnetization, one cannot assume a preferred age. Poor evidence for the age of acquisition would leave the palaeomagnetic studies of Palaeozoic rocks a muddle of ambiguous, inconsistent and ill-characterized data. Since a preferred interpretation cannot be assumed, geological situations must be found which provide 'full proof'.

Geological and physical criteria are used to interpret the evidence. In this particular instance, the field evidence includes: (1) rock textures, (2) micro-structural tests (Gillett, 1983), (3) regional tectonic history, (4) probable burial depths and temperature reached, (5) comparison of magnetic signature in ancient rocks, (6) the marked influence of PEF, (7) the nature of magnetic memory and behaviour of isolated component(s) in the course of thermal and AF demagnetization, and (8) the position of a component isolated with respect to the Curie point.

In the next section, arguments based on geological evidence and laboratory experiments are advanced to constrain the age of magnetization.

5.1. Evidence for a primary magnetization (PM)

Rock textures and microfolds capable of proving or disproving a primary magnetization are absent in the Brigus Shale Formation. Arguments in favour of PM include: (1) absence of metamorphism and consequently low temperatures of crystallization; (2) relatively shallow burial depth (2–3 km) as estimated from the reconstituted stratigraphic column; (3) probable absence of local rotation; (4) presence of haematite with unblocking temperatures up to 660 °C (theoretical blocking curves suggest that remagnetization could not occur over this geologic time-span); (5) local protection from major deformation. Although the gentle dip of these Cambrian strata is generally attributed to post-early Ordovician folding (possibly late Palaeozoic), McCartney (1967, 1969) believes that the Cambrian and Ordovician rocks in southeastern Conception Bay had an unusual protection from post-Ordovician deformation; the two sheltering factors being the underlying massive granitic basement and local Precambrian faults which offered pre-existing surfaces of movement for post-Cambrian faulting or post-Ordovician disturbances.

5.2. Evidence for a secondary magnetization (SM)

Secondary magnetization evidence of a lithological, tectonic and magnetic nature include the following. (1) Haematite-bearing red beds make good aquifers (Walker et al., 1981; Gillett, 1982; Larson and Walker, 1982; Larson et al., 1982) and their ability to retain a penecontemporaneous remanence is weak. (2) Techniques like vitrinite reflectance and conodont CAIS, used to obtain burial temperature, have not yet been applied to the Brigus Shale Formation; the estimated depth of burial based on stratigraphic grounds suggests maximum temperatures of 200 °C (geothermal gradient of 1 °C/30 m). Thermal remagnetization does not require deformation; heating from deep burial is sufficient. The sedimentary rocks of the Desert Range, Nevada, are only mildly deformed and metamorphosed; however, they are completely remagnetized (Gillett and van Alstine, 1982). The burial remagnetization may have been fossilized in the Brigus shale during tectonism at the close of the Palaeozoic, not because of major deformation, but rather because the section was uplifted and somewhat cooled at that time. (3) Magnetization with thermally distributed unblocking curves such as observed in the Brigus shale most commonly results from thermal or chemical remagnetization. Many such cases have been reported in Palaeozoic red-bed units from North America, e.g., Mauch Chunk Formation (Knowles and Opdyke, 1968), Bloomsburg Formation (Irving and Opdyke, 1965; Roy et al., 1967), Wood Canyon Formation (Gillett and van Alstine, 1979, 1982), Juniata Formation (van der Voo and French, 1977), Waynesboro Formation (Watts et al., 1980a) Catskill red beds (Kent and Opdyke, 1978; van der Voo et al., 1979; Miller and Kent, 1986a,b). Ancient primary red beds, when they yield a primary direction, generally do so in the last 10 or 15 °C before the Curie point, e.g., the Mississippian Mauch Chunk Formation restudied by Kent and Opdyke (1985). The direction reported in the original study of this formation (Knowles and Opdyke, 1968) was substantially steeper because of incomplete removal of the secondary component which had distributed blocking temperatures and was caused partially by heating from deep burial and chemical recrystallization. The primary component was isolated within 10 °C of the Curie point and had much shallower inclination. Thus, a thermally distributed, haematite-borne remanence of thermal and mainly chemical origin has been reported in geoclinal, Palaeozoic red beds throughout North America. A secondary remanence residing solely in haematite and resulting from protracted heating from burial could be expected in the Brigus shale given the geosynclinal nature of the section. (4) Given the low temperature corresponding to shallow burial, a secondary remanence of chemical origin is more likely. Data on the time–temperature relation for the magnetization of haematitic rocks suggest that if the origin of a red-bed component with an unblocking temperature of 650 °C were to be thermal, the rock unit must have suffered temperatures in excess of 600 °C. This situation does not hold for either the central Appalachians where the Mauch Chunk (Kent and Opdyke, 1985) crops out or for the Brigus shale of eastern Newfoundland. (5) The fact that magnetite is observed does not prove that haematite is not completely remagnetized. (6) The regional tectonic history and the agreement of the in situ Brigus directions with younger directions in the region (including some from plutonic rocks) suggest pervasive remagnetization. (7) The tilt test may not necessarily constrain the magnetization. The time of faulting, and therefore the attitudes resulting therefrom, may be younger than Cambrian. (8) It is noteworthy that fine-grained haematite appears to be remagnetized more easily than one might suspect from first-order theories (e.g., Pullaiah et al., 1975). A red member of the Wood Canyon Formation (Gillett and van Alstine, 1982) that is rich in fine-grained haematite has unblocking temperatures ranging up to > 650 °C. Theoretical blocking curves indicate that such a magnetization cannot occur, even over 550 Ma, except very near the Curie point. Still, the magnetization is late Palaeozoic in age. The resetting probably results from recrystallization in conjunction with thermal effects (van der Voo and Channell, 1980; Gillett and van Alstine, 1982). A thermochemical remagnetization of the Brigus shale is quite likely. (9) The univectorial vector paths observed in the Brigus shale can be explained

through a remagnetization process. Haematite blocking temperature spectra, especially those from secondary magnetizations, commonly overlap. Spurious linear paths may result from proportional removal of more than one component (French and van der Voo, 1977); this may explain the univectorial trajectories of the Brigus shale.

In summary, there are two alternatives for the magnetization of the Brigus shale (primary and secondary magnetizations). The evidence for a secondary magnetization (remagnetized component BR′) is better documented than for a penecontemporaneous origin for the characteristic component (BR).

5.3. Time of acquisition of magnetization

At least three ages of magnetization are plausible: (1) rotated primary magnetization (Lower Cambrian), (2) middle–late Devonian Acadian remagnetization, (3) post-folding remagnetization associated with the Permo-Carboniferous Alleghenian orogeny. The weak positive fold test and an inclination similar to that of the three late Precambrian rock units from Newfoundland characterized by a probable primary magnetization (Irving and Strong, 1985) favour a Lower Cambrian magnetization of the Brigus shale. Numerous geological and palaeomagnetic arguments favour a Devonian secondary magnetization. When freed of Kiaman overprinting, the palaeomagnetic data from Avalon have similar inclination. The St. Lawrence granite, the Garnish, Famine Back and Calmer rock units (Irving and Strong, 1985) and the Wesleyville dykes (Murthy, 1983) all have inclinations of ~ 50° and their age varies from 620 to 360 Ma. This points to Acadian overprinting. A slight improvement in grouping at 80% tilt correction suggests possible synfolding magnetization in Devonian times. Indeed, if we allow that some of the bedding tilt was acquired in the Acadian orogeny, then the slight improvement in grouping at 80% tilt correction strengthens the argument that the magnetization of the Brigus shale is Acadian in age. Considering the arguments presented, the second alternative appears to be the most acceptable and the first alternative is more probable than the last one.

6. Comparison with other palaeomagnetic data

The position of the BR pole depends on the choice of the polarity of the characteristic magnetization and on the amount of rotation of its declination. Assuming a negative inclination, and the absence of rotation of BR around a vertical axis, the BR pole is located at 27°S, 51°E. This pole lies far away from the younger (late Ordovician–Carboniferous) Palaeozoic poles (table 2 in Rao et al. (1986)) suggesting that we are dealing with older magnetization (Fig. 4). There are only three Cambrian poles reported for the Avalon: (1) pole CB (37°N, 176°E) derived from Cape Breton granites, the radiometric age (580–520 Ma) of which covers the entire span of the Cambrian (Rao et al., 1981), (2) pole MR (20°N, 146°W) derived from the steeply dipping early Cambrian Morrison River Formation, also from Cape Breton (Rao et al., 1986) and (3) pole BG (30°S, ~ 150°E) for the steeply dipping Middle Cambrian Bourinot Group (Johnson and van der Voo, 1985). The pole CB is derived from the older generation of granitic rocks (580–520 Ma, Cormier, 1973) which pre-date the early Cambrian Morrison River Formation; it is important to note in this context, that the abundant granitic plutonism present in Cape Breton Island corresponds to the Avalonian orogeny (600–500 Ma, Cormier, 1973). In view of the possible presence of a distributive type thermoremanence (in contrast to a discrete type) associated with long-duration intrusive activity (at least 50 Ma) in Cape Breton granites, it is likely that the pole CB corresponds to a late Cambrian age of magnetization. The authors believe therefore, that the CB magnetization is relatively younger Cambrian, compared with the MR magnetization obtained from the early Cambrian Morrison River Formation.

However, the BR (north) pole is not coincident with but falls in the vicinity of the MR (south) pole which could suggest a similar but perhaps not fully equivalent age of magnetization acquired some time during the early Cambrian. This is also in accordance with extensive trilobite faunal comparisons on Cambrian stratigraphy in southeastern Newfoundland, made by Hutchinson (1962) who considered the lower and upper parts of the

Brigus as probable equivalents to the Morrison River and MacCodrum Formations (from which the pole MR is derived), respectively. Moreover, comparison of the poles before (for the steeply dipping Morrison and MacCodrum Formations, pole MR′ at 2°S, 39°W with dp, dm = 6, 8°; for gently dipping Brigus red shale, pole BR′ at 25°S, 67°E with dp, dm = 10°, 14°) and after (correspondingly pole MR at 20°N, 146°W with dp, dm = 3°, 6°; pole BR at 17°S, 51°E with dp, dm = 6°, 9°) tilt correction indicates a better agreement in poles before rather than after the tilt correction is applied (Fig. 4) and suggests that the corresponding magnetizations for both formations post-date the tilting (possibly post-Ordovician).

Irving and Strong (1985) have reported palaeomagnetic results from three late Precambrian (623–606 Ma) rock units of the Marystown Group from Burin Peninsula, eastern Newfoundland. The three rock units (Calmer purple mudstones, Famine Back Cove basalts and Garnish red sandstones) form part of a shallowly dipping conformable subaerial volcanic–sedimentary sequence. The corresponding palaeomagnetic poles CM (32°S, 50°E), FB (1°S, 162°E) and GR (5°N, 148°E) are shown (Fig. 4). While the poles CM and BR (27°S, 51°E) are almost coincident and obtained from identical magnetizations pointing westerly with moderate downward inclinations, the other two poles FB and GR are derived from southeasterly magnetizations and lay further east from the CM and BR poles. Since the corresponding inclinations of the three rock units CM, FB and GR (after correction for tilting, which occurred possibly during the Acadian orogeny) are the same, Irving and Strong (1985) have interpreted the declination discrepancy in terms of possible vertical axis rotations during Acadian orogeny. Whether the similarity between the poles CM and BR is a consequence of contemporaneity in magnetization or the result of shear-related rotation, cannot be judged without additional palaeomagnetic data. However, the presence of dominant magnetization pointing westerly with moderate downward inclinations in three different sedimentary formations (MR, BR and CM) is not reported in the published data from the Avalon Zone in rocks younger than Cambrian.

Still, the assumption of a Cambrian age for pole BR is confronted by conflicting evidence. In particular, there is a noted inconsistency between the results of this study and those of Irving and Strong (1985). These authors suggested a rotation of the formations around a vertical axis, while the results of this study do not. In addition, the Cambrian pole position obtained by Johnson and van der Voo (1985) for Avalon is significantly different from pole BR. Finally, if a primary magnetization is assumed (north: 27°N, 129°W) for pole BR close to MR and the corresponding APW path suggested is correct, then Avalon would have had to have been in the northern hemisphere during the Cambrian, or alternatively Avalon must have undergone a 180° rotation sometime between the early Cambrian and Devonian (if the alternate polarity is used). Based on what is known about the location of Gondwana, the probable home of Avalon, at this time (near the South Pole), the first alternative is unlikely. The second, though possible, seems improbable. For the above-mentioned reasons, pole BR′ resulting from a secondary magnetization is most likely.

In the absence of a confirmatory fold test, since it is not possible to provide evidence for the primary nature of magnetization, the other likely and preferred alternative is to consider that the magnetization is post-tilting. In this case the in situ remanence direction is 266°, +58° with a corresponding palaeomagnetic (south) pole BR′ located at 25°N, 113°W (dp, dm = 10°, 14°) (Table I, Fig. 4). This is not significantly different from poles CB2 (Cape Breton granites: Rao et al., 1981), JA (St. George gabbro, New Brunswick: Roy et al., 1979) and JF (Mascarene Group siltstones, New Brunswick: Roy and Anderson, 1981) the radiometric ages of which roughly average around 400 Ma. It is noteworthy that the uncorrected pole (BR′) falls in the grouping of directions approximately between CB2 and CB3. Since the corrected BR pole falls into the tail of directions from Cambrian rocks (MR, CB, BG, FB and GR), it may very well be that the tail could be an artifact of applying an inappropriate structural correction to remagnetizations. This interpretation appears more correct than explaining the tail by postulating large APW from the Cambrian to

Ordovician. It is particularly intriguing that the uncorrected BR′ direction is located near the St. George gabbro (JA) pole. The presence of such plutonic rocks in New Brunswick and Nova Scotia (presumably resulting from the Acadian orogeny) surely suggests that geothermal gradients could have been elevated, which would facilitate thermal remagnetization of the Brigus shale. Consequently, the uplift that could lock-in a thermo-chemical remagnetization, by raising and cooling the section, could follow such tectonic activity. The thermal process is accompanied by chemical magnetization (aqueous reduction and dissolution of red pigment).

Therefore, based mainly on the thermally distributed characteristic magnetization and to a lesser extent on the comparison with other Avalon Palaeozoic palaeopole positions, the Brigus shale is most likely to reflect Siluro-Devonian remagnetization.

7. Tectonic significance of Avalon results

Tectonic significance of the Avalon Zone is well described by Wilson's (1966) model of an early Palaeozoic proto-Atlantic Ocean, which is amenable to palaeomagnetic tests (Deutsch, 1969, 1980, 1984). (1) The results can be used to compare contemporaneous Cambrian data obtained from the North American craton and the Avalon suspect terrane. The apparent scatter and complexity of the cratonic Cambrian data (Watts et al., 1980b; Seguin et al., 1981) suffers from a lack of good control on the age of magnetization. However, the BR pole is significantly different from most of the cratonic Cambrian poles; this could suggest that Avalon was a displaced terrane with respect to the North American craton during the Cambrian. *Assuming a primary magnetization*, conservative estimates of palaeolatitudes for some of the cratonic Cambrian data reveal a difference of at least 15° with respect to the Brigus shale ($p = 29 \pm 6°$), suggestive of a displaced nature of the Avalon with respect to the craton. (2) The results can be used to compare Cambrian data from crustal fragments that once formed the eastern margin of the hypothetical proto-Atlantic Ocean during the early Paleozoic. However the reliability of available Cambrian data from Europe (Duff, 1980; Perroud et al., 1982; Burrett, 1983) is not adequate. (3) The results can be used for across-suture comparison within Newfoundland, i.e. a comparison of contemporaneous Cambrian data obtained from western Newfoundland, which is considered part of cratonic North America from both geologic (Williams, 1980) and palaeomagnetic (Deutsch and Rao, 1977) evidence, with that from the Avalon Peninsula in eastern Newfoundland. The only reported Cambrian pole from western Newfoundland is from the basal Cambrian Bradore red sandstones (Rao and Deutsch, 1976).

If the Brigus shale Formation is remagnetized, as is most probably the case, its pole cannot be used to define the apparent polar wander path (APWP) of the Avalon Zone during Cambrian times or to calculate the opening of the Iapetus, but a minimum amount of opening can be estimated. In spite of the limited palaeomagnetic data, a preliminary attempt to outline the APWP of Avalon is a worthwhile exercise. From the presently published data the following features are apparent in Fig. 4: (1) a probable sequence of Cambrian poles from MR to CB and BG; (2) late Ordovician poles DP (van der Voo and Johnson, 1985) and CB1 (Rao et al., 1981); (3) Siluro-Devonian poles JA, JF, (Roy et al., 1979), CB2 (Rao et al., 1981) and BR′ (this study); (4) a group of early Devonian poles JD, JE, CB3, WA (see discussion by Rao et al., 1981; Seguin et al., 1981); and (5) a group of Carboniferous poles TV and TH (Kent, 1982), MS (Roy and Park, 1974), SH and SR (Irving and Strong, 1985). A reliable APWP for the Avalon terrane awaits additional early Palaeozoic palaeomagnetic results. Considering the paucity of the palaeomagnetic data for Avalon, it would be premature to draw conclusions on intercontinental correlations. However, by Middle Devonian times, both Avalon and the North American craton poles coincide, implying that Avalon docked on to North America during that period.

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